

U.2 PCIe Gen4 SSD

Version 1.2
Dec. 02, 2025

TD7P40

3840GB, 7680GB

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Revision History

Revision	Date	Description	Editor
1.0	May. 05, 2025	Initial release	
1.1	Nov. 04, 2025	Modify Ordering Information	
1.2	Dec. 02, 2025	Modify Model Number, Environmental Specifications, Electronic Specification	

Form Factor	Model Number	Capacity	LBA Count (512B Size)
U.2	TRUSTA TD7P40-003T8T5U2022D020	3840GB	7,501,476,528
	TRUSTA TD7P40-007T6T5U2022D020	7680GB	15,002,931,888

Features	Environmental Specifications	
PCI Express Gen4 x 4 Lane (Single Port)	Temperature Range	
PCI Express Base Specification Rev. 4.0	- Operating (by SMART)	0°C ~ 70°C
NVM Express Specification Rev. 1.4	- Non-Operating	-40°C ~ 85°C
Power Loss Protection (by On-board Circuit)	Shock (Non-Operating, Half Sine)	1500G / 0.5ms
Thermal Throttling	Vibration	
SMBus Basic Management (Out of Band)	- Non-Operating, Sine	20G / 20 ~ 2000Hz
Secure Boot, Crypto Erase	- Non-Operating, Random	3.13Grms / 5 ~ 800Hz
Security AES GCM 256bits, SHA2-512, RSA-4096	- Operating, Random	1.5Grms / 5 ~ 500Hz
TCG Opal SSC v2.01	Humidity (Non-Operating)	5 ~ 95%

Drive Configuration		Reliability Specifications	
Interface	PCIe Gen4x4	UBER	< 1 Error per 10 ¹⁷ Bits Read
NAND	SanDisk BiCS5 eTLC	MTBF	2,000,000 hours
Protocol	NVMe v1.4	Endurance	1.0DWPD (5 Years Assumed)
Sector Size	512B, 512B+8B, 4KiB, 4KiB+8B, 4KiB+64B		
		Data Retention	3 Months (40°C)

Performance (Typ.)		Quality of Service (Latency)	
Data Transfer Rate (128KB Data Size, Up to)		Average Latency (4KB Data Size, Sustained, QD1, Typical)	
Sequential Read	Up to 7400 MB/s	Random Read	72 us
Sequential Write	Up to 5050 MB/s	Random Write	12 us
Data I/O Speed (4KB Data Size, Sustained, Up to)			
Random Read	Up to 1300K IOPS		
Random Write	Up to 200K IOPS		

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1.0 Overview

1.1 General Description

The TD7P40 is an NVMe v1.4-compatible SSD featuring a PCIe Gen4 x4 interface in the SFF-8639 (U.2) form factor. It delivers high performance and consistent Quality of Service (QoS), along with world-class reliability for PCIe NVMe-based systems. Available in 3840GB and 7680GB capacities, the TD7P40 is well-suited for performance-driven and mission-critical storage applications.

1.2 Product List

Following table has U.2 2280 form factor SSD model numbers with 2 capacities.

Type	Model Number	Capacity
U.2	TRUSTA TD7P40-003T8T5U2022D020	3840GB
	TRUSTA TD7P40-007T6T5U2022D020	7680GB

Table 1-1 U.2 Products

2.0 Mechanical Specifications

2.1 Physical Dimensions and Weights

Capacity(GB)	Length(mm)	Width(mm)	Height(mm)	Weight(gram)
3840	100.45 - 0.25	69.85 ± 0.15	15.00 +0.10, -0.40	
7680	100.45 - 0.25	69.85 ± 0.15	15.00 +0.10, -0.40	

Table 2-1 Physical Dimension and Weight

2.2 Mechanical Drawing

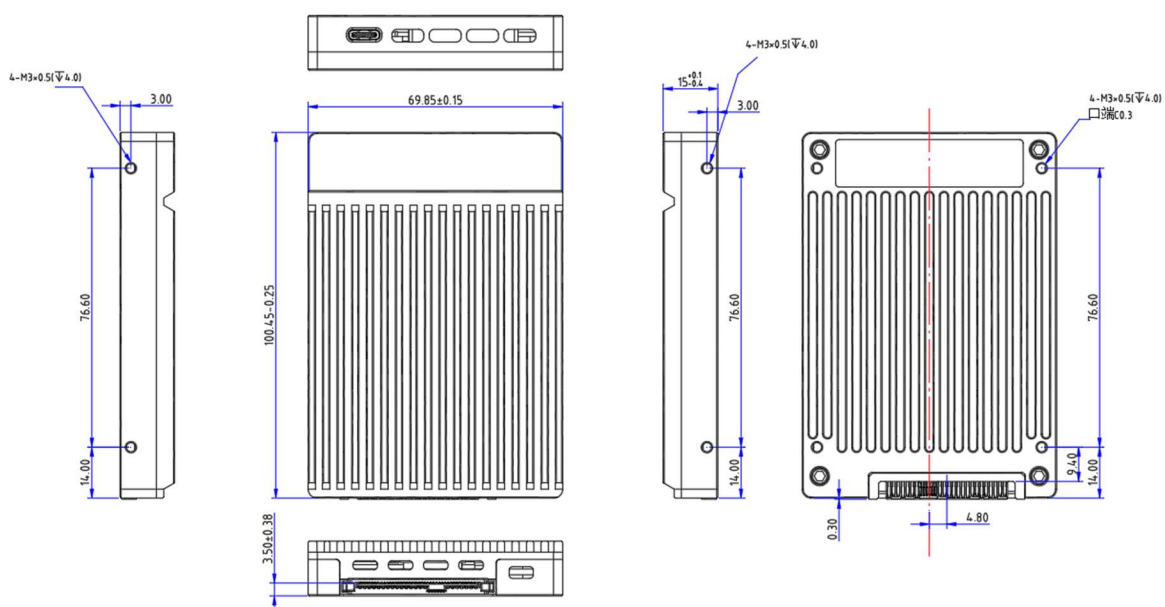


Figure 2-1 Product Dimensions

3.0 Product Specifications

3.1 Interface Specification

- Compliant with PCI Express U.2 Specification Revision 1.1.
- Compliant with NVMe 1.4 register interface and command set.
- Compliant with PCIe Express 4.0.

Top Side			Bottom Side		
NO.	Pin	Descriptions	Descriptions	Pin	NO.
75	GND	System Ground			
73	GND	System Ground	+3.3V	POWER	74
71	GND	System Ground	+3.3V	POWER	72
69	PDEDC	PDEDC	+3.3V	POWER	70
67	NC	NC	SUSCLK	NC	68
M-KEY					
57	GND	System Ground	MFG_CLK	NC	58
55	Diff	RefCLKP	MFG_DATA	NC	56
53	Diff	RefCLKN	PEWAKE#	NC	54
51	GND	System Ground	CLKREQ#	CLKREQ	52
49	Diff	P0RXP	PERST#	PERST	50
47	Diff	P0RXN	NC	NC	48
45	GND	System Ground	NC	NC	46
43	Diff	P0TXP	ALERT# (Optional)	ALERT	44
41	Diff	P0TXN	SMB_DATA (Optional)	I2C	42
39	GND	System Ground	SMB_CLK (Optional)	I2C	40
37	Diff	P1RXP	NC	NC	38
35	Diff	P1RXN	NC	NC	36
33	GND	System Ground	NC	NC	34
31	Diff	P1TXP	NC	NC	32
29	Diff	P1TXN	NC	NC	30
27	GND	System Ground	NC	NC	28
25	Diff	P2RXP	NC	NC	26
23	Diff	P2RXN	NC	NC	24
21	GND	System Ground	NC	NC	22
19	Diff	P2TXP	NC	NC	20
17	Diff	P2TXN	+3.3V	POWER	18
15	GND	System Ground	+3.3V	POWER	16
13	Diff	P3RXP	+3.3V	POWER	14
11	Diff	P3RXN	+3.3V	POWER	12
9	GND	System Ground	LED1#(OPTION)	INDICATE	10
7	Diff	P3TXP	NC	NC	8
5	Diff	P3TXN	NC	NC	6
3	GND	System Ground	+3.3V	POWER	4
1	GND	System Ground	+3.3V	POWER	2

Table 3-1 Pin assignment and descriptions

3.2 Capacity

OP	Drive Capacity	LBA Count
OP7	3840GB	7,501,476,528
	7680GB	15,002,931,888

Table 3-2 LBA Count for Each Capacity

3.3 Performance

To measure the performance, following test environment was used. These platforms were used to measure sequential/random performance, latency.

Mainboard	ASUS PRIME Z690-A
CPU	Intel i9-12900K
DRAM	64GB
O/S	Ubuntu 20.04.5 LTS
I/O Engine	Linux native driver and Kyber I/O scheduler
Test Tool	FIO (3.35)

Table 3-3 Workstation Specification

3.3.1 Sequential Performance

Performance	3840GB	7680GB	Unit
Sequential Read	7300	7400	MB/s
Sequential Write	2700	5050	MB/s

Table 3-4 Sequential Performance

3.3.2 Random Performance

Performance	3840GB	7680GB	Unit
4K Random Read	1200K	1300K	IOPS
4K Random Write	145K	200K	IOPS

Table 3-5 Random Performance

3.3.3 IOPS Consistency

Performance	3840GB	7680GB
Random Read (4KB)	> 99%	
Random Write (4KB)	> 99%	

Table 3-6 IOPS Consistency

NOTE:

IOPS consistency (%) = (99.9% IOPS) / (Average IOPS) x 100. (Queue Depth = 32)

3.3.4 Latency

Performance	1920GB	3840GB	Unit
Random Read Latency	72		us
Random Write Latency	12		us

Table 3-7 Average Latency

NOTE:

1) Thread Count = 1, Queue Depth = 1, IO Size = 4KB

3.3.5 Power on Ready Time

Description	1920GB	3840GB	Unit
After Normal Shutdown	6		sec
After Unsolicited Shutdown	20		sec

Table 3-8 Power on Ready Time per Shutdown Types

3.4 Electrical Specifications

3.4.1 Operating Voltage

Operating Voltage	
Input Power	DC 12V ± 10%
Maximum Allowable Ripple	100mV p-p

Table 3-9 Operating Voltage

3.4.2 Operating Current

Description	3840GB	7680GB	Unit
Peak Current	2.05	2.05	A

Table 3-10 Operating Current

-Peak Current is the maximum operating current during a test with 16 μs sampling rate and 100 μs sampling period.

3.4.3 Power Consumption

Description	3840GB	7680GB	Unit
Read/Write	13	19	W
Idle	3	3	W

Table 3-11 Power Consumption (Typical)

- The typical value means to measure the power consumption by using IO Meter with 128KB Sequential and 4K Random read/write transfers within 15 minutes.
- The measurement may vary among different host systems and settings.

3.5 Environmental Conditions

Feature	Operating	Non-Operating
Standard Temperature	0°C to 70°C	-40°C to 85°C
Humidity	5%~95% RH, non-condensing	

Table 3-12 Temperature and Humidity

3.6 Reliability

Reliability	Values	Condition
UBER	$\leq 10^{-17}$	
MTBF	2 million hours	
Power On/Off	50,000	
Data Retention	3 months	Temperature $\leq 40^{\circ}\text{C}$

Table 3-13 Reliability

Parameter	3840GB	7680GB
DWPD	1	1

Table 3-14 DWPD

3.7 Endurance

Capacity	3840GB	7680GB	Unit
TBW	7,008	14,016	TB

Table 3-15 Endurance

4.0 Support Command Sets

4.1 Command Set

Command Set	Command	Note
Admin Command Set	Delete I/O Submission Queue	supported
	Create I/O Submission Queue	supported
	Get Log Page	supported
	Delete I/O Completion Queue	supported
	Create I/O Completion Queue	supported
	Identify	supported
	Abort	supported
	Set Features	supported
	Get Features	supported
	Asynchronous Event Request	supported
	Namespace Management	supported
	Firmware Commit	supported
	Firmware Image Download	supported
	Device Self-test	supported
	Namespace Attachment	supported
	Format NVM	supported
	Security Send	supported
	Security Receive	supported
	Sanitize	supported
I/O Command Set	Flush	supported
	Write	supported
	Read	supported
	Write Uncorrectable	supported
	Compare	supported
	Write Zeroes	supported
	Dataset Management	supported
	Verify	supported

4.1.1 Identify Device Command

IDENTIFY DEVICE (ECh). These commands read out 512Bytes of drive parameter information. Parameter Information consists of the arrangement and value as shown in the following table. This command enables the host to receive the Identify Drive Information from the device.

Bytes	Field	Description	Value
1:0	VID	PCI Vendor ID	1CC1
3:2	SSVID	PCI Subsystem Vendor ID	1CC1
23:4	SN	Serial Number (ASCII)	Variables
63:24	MN	Model Number (ASCII)	Variables ex) TRUSTA TD7P40-007T6T5U2022D020
71:64	FR	Firmware Revision	Variables
72	RAB	Recommended Arbitration Burst	2h
75:73	IEEE	IEEE OUI	707C18
76	CMIC	Controller Multi-Path I/O and Namespace Sharing Capabilities Bit 3: 0b - not support Asymmetric Namespace Access Reporting Bit 2: 0b - Controller is associated with a PCI Function. Bit 1: 0b - Device has a single controller Bit 0: 0b - Device has a single NVM subsystem port	0h
77	MDTS	Maximum Data Transfer Size 9h: 2MB	9h
79:78	CNTLID	Controller ID (CNTLID)	1h
83:80	VER	Version	00020000h
87:84	RTD3R	RTD3 Resume Latency	1C9C380h
91:88	RTD3E	RTD3 Entry Latency	4C4B40h
95:92	OAES	Optional Asynchronous Events Supported Bit 9: 1b - supports the Firmware Activation Notices event. Bit 8: 1b - supports the Namespace Attribute Notices event and the associated Changed Namespace List log page.	300h
99:96	CTRATT	Controller Attributes Bit 19: 1b – supports the Flexible Data Placement capability (refer to TP4146 Flexible Data Placement 2022.11.30 Ratified.pdf) Bit 9: 1b – supports reporting of a UUID List Bit 7: 1b - supports reporting of Namespace Granularity	00280h
101:100	RRLS	Read Recovery Levels Supported	0h
111	CNTRLTYPE	Controller Type 1h – I/O Controller	1h
127:112	FGUID	FRU Globally Unique Identifier	”00000000000000000000000000000000”
129:128	CRDT1	Command Retry Delay Time 1	0h
131:130	CRDT2	Command Retry Delay Time 2	0h
133:132	CRDT3	Command Retry Delay Time 3	0h
257:256	OACS	Optional Admin Command Support Bit 5: 1b – support Directives	5Fh

		Bit 3: 1b - support the Namespace Management capability Bit 2: 1b - support the Firmware Commit and Firmware Image Download Bit 1: 1b - support Format NVM Bit 0: 1b - support Security Send and Receive	
258	ACL	Abort Command Limit (zero based value) (Maximum number of concurrently outstanding Abort commands)	3h
259	AERL	Asynchronous Event Request Limit (zero based value) (Maximum number of concurrently outstanding Asynchronous Event Request commands)	Fh
260	FRMW	Firmware Updates Bit 4: 1h - support firmware activation without a reset Bits 3:1 - Number of firmware slots Bit 0: 0h - Slot 1 is read/write	14h
261	LPA	Log Page Attributes Bit 3: 1h - support the Telemetry Host-Initiated and Telemetry Controller-Initiated log pages and sending Telemetry Log Notices Bit 2: 1h - support extended data for the Get Log Page Command Bit 1: 1h - support the Command Supported and Effects log page	1Eh
262	ELPE	Error Log Page Entries (Number of Error Information log entries stored by controller)	3Fh
263	NPSS	Number of Power States Support (zero based value)	4h
264	AVSCC	Admin Vendor Specific Command Configuration Bit 0: 1h - all Admin Vendor Specific Commands use the format defined in nvme specification	1h
265	APSTA	Autonomous Power State Transition Attributes (APSTA) 0h: not support	0h
267:266	WCTEMP	Warning Composite Temperature Threshold 15Eh: 77°C	15Eh
269:268	CCTEMP	Critical Composite Temperature Threshold 166h: 85°C	166h
271:270	MTFA	Maximum Time for Firmware Activation 0h: undefined	30
275:272	HMPRE	Host Memory Buffer Preferred Size 0h: not support	0h
279:276	HMMIN	Host Memory Buffer Minimum Size 0h: not support	0h
295:280	TNVMCAP	Total NVM Capacity	- 3840GB: 3840755982336 Bytes - 7680GB: 7681501126656 Bytes
311:296	UNVMCAP	Unallocated NVM Capacity	0h
315:312	RPMBS	Replay Protected Memory Block Support 0h: not support	0h
317:316	EDSTT	Extended Device Self-test Time 0h: not support	Ch
318	DSTO	Device Self-test Options 0h: not support	1h
319	FWUG	Firmware Update Granularity 1h: 4KB	1h
321:320	KAS	Keep Alive Support 0h: not support	0h

323:322	HCTMA	Host Controlled Thermal Management Attributes 0h: not support	0h
325:324	MNTMT	Minimum Thermal Management Temperature 0h: not support	0h
327:326	MXTMT	Maximum Thermal Management Temperature 0h: not support	0h
331:328	SANICAP	Bit 31-30: 01b - Media is not additionally modified by the NVMe controller after sanitize operation completes successfully Bit 29: 0b - supports the No-Deallocate After Sanitize bit in a Sanitize command Bit 2: 1b - supports the Overwrite sanitize operation Bit 1: 1b - supports the Block Erase sanitize operation Bit 0: 1b - supports the Crypto Erase sanitize operation	0x400000007h
335:332	HMMINDS	Host Memory Buffer Minimum Descriptor Entry Size	0h
337:336	HMMAXD	Host Memory Maximum Descriptors Entries	0h
339:338	NSETIDMA X	NVM Set Identifier Maximum	0h
341:340	ENDGIDMA X	Endurance Group Identifier Maximum	0h
342	ANATT	ANA Transition Time	0h
343	ANACAP	Asymmetric Namespace Access Capabilities	0h
347:344	ANAGRPM AX	ANA Group Identifier Maximum	0h
351:348	NANAGRPI D	Number of ANA Group Identifiers	0h
355:352	PELS	Persistent Event Log Size 18h: 1.5MB	18h
512	SQES	Submission Queue Entry Size 66h: Max 64B / Min 64B	66h
513	CQES	Completion Queue Entry Size 44h: Max 16B / Min 16B	44h
515:514	MAXCMD	Maximum Outstanding Commands 0h: no limit	0h
519:516	NN	Number of Namespaces	- 3840GB: 40h - 7680GB: 80h
521:520	ONCS	Optional NVM Command Support Bit 7: 0h - not support the Verify Bit 6: 1h - support the Timestamp feature Bit 5: 0h - not support Reservations Bit 4: 1h - support the Save field in the Set Feature & the Select field in the Get Feature Bit 3: 1h - support the Write Zeros Bit 2: 1h - support the Dataset Management Bit 1: 1h - support the Write Uncorrectable Bit 0: 0h - not support Compare	5Eh
523:522	FUSES	Fused Operation Support Bit 0: 0h not support	0h
524	FNA	Format NVM Attributes Bit 2: 1h - support cryptographic erase as part of the secure erase Bit 1: 0h - support the secure erase per namespace basis Bit 0: 0h - support the Format per namespace basis	4h
525	VWC	Volatile Write Cache Bit 2:1 - The Flush command supports the NSID field set to FFFFFFFFh Bit 0: 0h - No VWC present	6h

527:526	AWUN	Atomic Write Unit Normal (zero based value)	depending on namespaces (Default=0)
529:528	AWUPF	Atomic Write Unit Power Fail (zero based value)	depending on namespaces (Default=0)
530	ICSVSCC	I/O Command Set Vendor Specific Command Configuration Bit 0: 1h - all NVM Vendor Specific Commands use the format defined in nvme specification	1h
531	NWPC	Namespace Write Protection Capabilities	0h
533:532	ACWU	Atomic Compare & Write Unit 0h – not support	0h
535:534	CDFS	Copy Descriptor Formats Supported Bit 0: 0h does not support Copy Descriptor Format 0h	0h
539:536	SGLS	SGL Support - not support SGL	0h
543:540	MNAN	Maximum Number of Allowed Namespaces	0h
1023:768	SUBNQN	NVM Subsystem NVMe Qualified Name	
529:528	AWUPF	Atomic Write Unit Power Fail (zero based value)	depending on namespaces (Default=0)

Table 4-1 Identify Device Table

Notes:

F/V = Fixed/variable content.

F = the content of the word is fixed and does not change. For removable media devices, these values may change when media is removed or changed.

V = the contents of the word is variable and may change depending on the state of the device or the commands executed by the device.

X = the content of the word may be fixed or variable.

4.2 SMART Health Information

The following table defines the vendor specific data in byte 2 to 361 of the 512-byte SMART data.

Bytes	Default	Description
0	0	Critical Warning • Bit 7:5 – Reserved • Bit 4 – 1h: the volatile memory backup device has failed. (only valid if the controller has a volatile memory backup solution) • Bit 3 – 1h: the media has been placed in read only mode • Bit 2 – 1h: the device reliability has been degraded due to significant media related errors or any internal error that degrades device reliability • Bit 1 – 1h: the temperature has exceeded a critical threshold • Bit 0 – 1h: the available spare space has fallen below the threshold
2:1	Current temp	Temperature
3	100	Available Spare
4	10	Available Spare Threshold
5	0	Percentage Used
6	0	Endurance Group Critical Warning Summary
31:7	-	Reserved
47:32	0	Data Units Read
63:48	0	Data Units Written
79:64	0	Host Read Commands
95:80	0	Host Write Commands
111:96	0	Controller Busy Time
127:112	0	Power Cycles
143:128	0	Power On Hours
159:144	0	Unsafe Shutdowns
175:160	0	Media Errors
191:176	0	Number of Error Information Log Entries
511:192	-	Reserved
195:192	0	Warning Composite Temp Time
199:196	0	Critical Composite Temp Time
201:200	Current temp	Temp Sensor 1
203:202	Current temp	Temp Sensor 2
205:204	Current temp	Temp Sensor 3
511:204	0	Reserved

Table 4-2 S.M.A.R.T. Attribute

5.0 End-to-End Data Protection

Both SRAM and DDR are deployed with ECC protection, and feature 1-bit error correction and detection of two or more bits error.

CRC16 algorithm is used for each Sector to protect data transfer between host and NAND, supporting 512-Byte Sector Size. LBA number is used as the seed of CRC16, to avoid silent data corruption when returning incorrect Sector data.

6.0 Package Specification

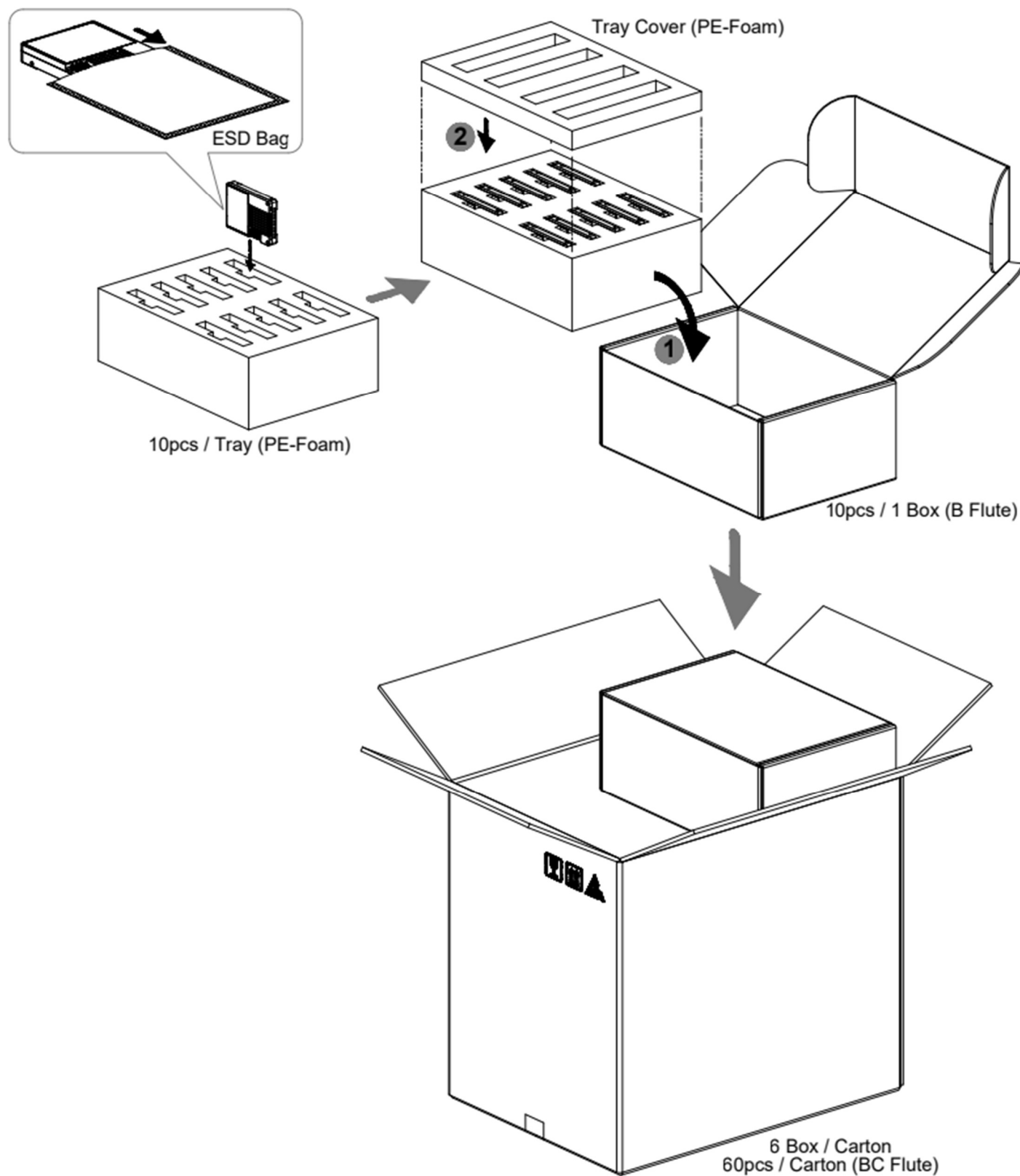


Figure 6-1 Package Specification

7.0 Ordering Information

Model Name	Capacity	DWPD	Type	Remark
TRUSTA TD7P40-003T8T5U2022D020	3840GB	1	U.2 PCIe Gen4	0°C~70°C
TRUSTA TD7P40-007T6T5U2022D020	7680GB	1	U.2 PCIe Gen4	

Table 7-1 Ordering Information